Various Methods for Predictive Analysis of Unknown Causes of Wildfires

1st Heidi Fox
University of Dayton
Dayton, United States of America
foxh06@udayton.edu

2nd Victor Nusekabel
University of Dayton
Dayton, United States of America
nusekabelv1@udayton.edu

3rd Claire Summers
University of Dayton
Dayton, United States of America
summersc3@udayton.edu

4th Alex Penzel
University of Dayton
Dayton, United States of America
penzela1@udayton.edu

***Abstract*—This document is a model and instructions for LaTeX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.**

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

According to data collected by researchers at the University of Maryland, recently updated to cover the years 2001-2024, forest fires burn more than twice as much tree cover each year as they did two decades ago [1]. That wildfires have been much worse in the past decades has become a fact widely accepted by science. Now, more than ever, it is important to develop predictive models for when, why, and where wildfires may occur. According to some sources, authorities often fail to identify the source of more than half of wildfires [2]. This leads to a huge information gap that lead to less accuracy in predictive models. It is important to attempt to understand what these unknown causes are to help those in charge of disaster mediation plan the allocation of resources.

The selected data set was sourced from Kaggle and titled "WiDS 2026 External Seasonal Wildfire Risk Features." The dataset contains just over 4000 rows of data corresponding to different instances of wildfires and their early signals, ranging from 1881 to 2025. This is a dataset intended for use in a data analytics competition. The original dataset only included instances of wildfires occurring in 1950 or after and was compiled from various open-source wildfire and climate archives (NOAA, NASA, FAO, GFED). The original, non-synthetic data makes up about 50% of the dataset. Data for wildfires prior to 1950 was synthetically generated but based on real data. This dataset adds long-term seasonal wildfire patterns to help models capture background risk conditions.

The data has ten attributes, three of which are categorical, and the other seven are numerical data. The categorical attributes are country, region, and cause of the fire. The possible causes listed for the fires are Human, Climate Change, Deforestation, Lightning, and Unknown. The numerical attributes are year, month, total number of fires, total burned area (hectares), average temperature (C), average relative humidity percentage, and average wind speed (km/h). There are 13 unique countries, which encompass 28 regions. Three of these regions are in the US: California, Oregon, and Texas. The totals and averages are considered over the given month and geographic region.

The purpose of the following analysis is to use predictive models to categorize the instances of wildfires with "unknown" causes into the four known causes. The following sections describe relevant research and development of similar models, the methodology used in this study, and the results and conclusions.

II. Related Research

Data Analytics is already being used to identify causes of wildfires to help understand what can be done as preventative measures. A study in the Journal of Forestry Research [3] used machine-learning-based max entropy models to help capture the randomness of wildfire occurrences and predict the spatial distribution of fires to determine risk zones for each cause. Another study conducted by the University of Seoul in 2026 [4] used "Explainable" Machine Learning Models in combination such as Random Forest, XGBoost, Light GBM, and logistic regression. The study focused on ignition causes such as distance to roads and agricultural land, which are traditionally treated as "independent variables" in most regression models. The study emphasized that human accessibility ignition factors interact with fire-weather. This confirms what other studies [5] have found: wildfires are becoming increasingly human-caused and human factors that could once be analyzed as independent can no longer be thought of as such.

With a wide variety of machine-learning models and complex algorithms available, identifying methods for data analysis is daunting due to the sheer number of options. A study conducted in 2023 [6] that used both Random Forest and Linear Regression actually found in general, Linear Regression had a better performance in predicting forest fires. The author of the paper noted that Linear Regression is "highly efficient for small datasets" and produces results that are "easy for practitioners to understand and use for decision-making." Another paper [7] utilized classical ordinary least squares (OLS) linear regression with geographically weighted regression (GWR) to understand long-term patterns in wildfire occurrences. The researchers concluded that models that make differentiations between regions have more explanatory power and help to provide insight into a large region, while also recognizing local characteristics. Simpler models like Linear Regression provide an essential baseline before further analysis is performed. Another analysis [8] was performed using regression trees and random forest analysis and was more focused on predicting which regions were most prone to wildfires. The dataset the study was based upon had similar attributes to those of the dataset used for this study. The study found that some truths held both globally and regionally: the greatest wildfire burned area was associated with high temperatures, intermediate annual rainfall, and prolonged dry periods (which did vary by region). Although the regions with the highest fire incidence did not demonstrate a systemic pattern. The study ultimately attributes this to "anthropogenic variables", meaning human factors and argues that in some cases these human factors outweigh climate factors.

These and other studies show that linear models often excel at demonstrating the relationship between "distance to roads" or "population density" and fire frequency because these relationships are often more linear than complex weather patterns. Most researchers note that while linear regression is great for identifying causes, it may have lower accuracy than non-linear models because weather variables (such as humidity vs. temperature) interact in complex ways [9].

For some problems, increased complexity is needed. In another study that focused on creating predictive models for wildfires, Neural Networks achieved 98.32% accuracy using the MLPClassifier Support Vector Machine achieved 97.48% accuracy using C-Support Vector Classifier (SVC) [10]. In this particular study the data was created based on remote sensing data, which other studies that have used remote sensing data have found success as well [9].

In a study centered in Northern California, the Random Forest model outperformed all other models used to predict wildfires [11]. This analysis used other variables like power lines, vegetation, and terrain to aid in the model's performance. This data was input into two different random forest classifiers, which were then used in a stacked generalization ensemble. These were considered to be the two "weak" models, creating the base model; they were then input into "the meta classifier model, resulting in a robust ensemble model."

A study in Spain [16] identified human factors associated with high forest risk and analyzed the spatial distribution of forest fires in the country. A binary model estimated the probability of high or low occurrence of forest fires. This was defined by an ignition danger index that is currently used by the Spanish forest service. The study utilized logistic regression to effectively understand human factors affecting fire ignition in Spain with 85.4% accuracy.

As the literature on the topic of wildfire predictions and analysis, determining the causes of fires is extremely important for the development of mitigation strategies and resource allocation. For this reason, using predictive models to "fill in the gaps" for what we may know about the cause of a wildfire is relevant research to the field.

III. Methods and Results

The dataset used for this analysis and the prediction of unknown causes of wildfires included over 4000 instances of wildfires globally. As mentioned in the previous section [5] [6] [7], different factors affect analysis in different regions, and there are complications associated with applying the same predictive model to China and Greece for example. For this reason, initially, the dataset was reduced to only instances of fires occurring in the U.S. This included three regions: California, Texas, and Oregon. Narrowing the dataset to just U.S. Data reduces the size of the dataset and the number of instances of wildfires to 435.

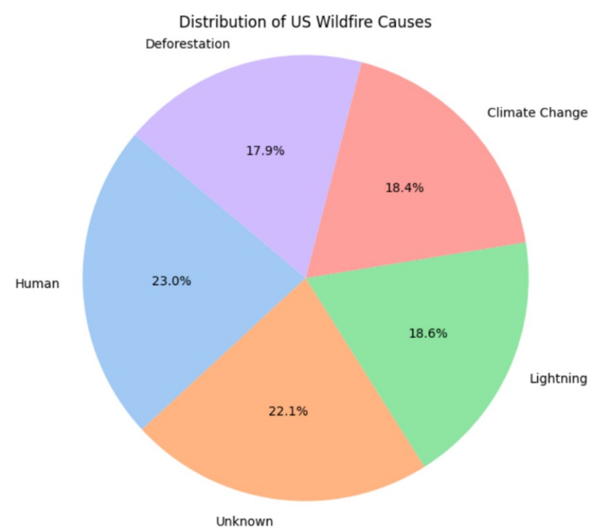

Fig. 1. Distribution of Causes of Wildfires Across US Wildfires

A. K-means Clustering on PCA

After performing Principal Component Analysis on the U.S. data, including the three regions and 5 types of causes, a K-means clustering algorithm was used on the resulting PCA chart in order to better visualize what the data looks like. What the K-means clustering algorithm did was try to identify and assign a centroid to groupings based on n-clusters given. After running the K-means algorithm on the visualization provided by PCA, the best parameters that could be applied and seen on the PCA was the clustering of the different regions that were present before. The algorithm was only good at creating

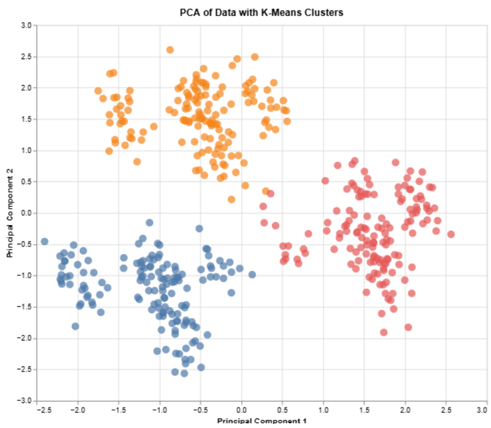

Fig. 2. K-Means Clustering on PCA

groups based off the regions of California, Texas, and Oregon due to the fact that they were the only groupings that had a well established centroid. Any other attempts at visualizing with other groups would split and separate groups that were together. This resulted in only three clusters being legible and only being useful towards the clustering.

B. DBSCAN Clustering on PCA

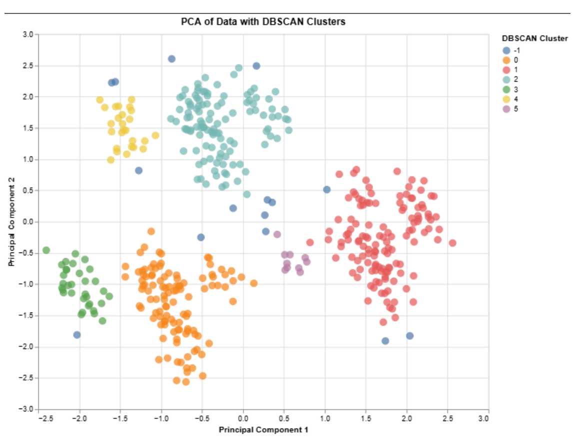

Fig. 3. DBSCAN Clustering on PCA

Another algorithm that was used on the resulting PCA was DBSCAN clustering which was used in order to help visualize the data better. What DBSCAN did was take in two hyperparameters in order to create a density area in order to cluster groups together that also included noise points. These two hyperparameters that helped the algorithm were the epsilon parameter and the number of samples. What the epsilon parameter did was set the distance between different points that would be counted in a cluster. If points were in the distance, they would be classified as being in the same cluster. The other parameter which was the number of samples helped the algorithm to determine the minimum number of samples that needed to be nearby in order to be counted in the cluster. What the DBSCAN algorithm gave was a visualization of causes that were grouped together. Due to overlay, some causes such as lightning, unknown, human, and climate change were grouped together due to their high density. Other groups such as deforestation was left out in its own density grouping. The last grouping that the algorithm gave was an outlier group that classified points that did not fit in with the other density clusters.

Performing clustering algorithms on PCA, as well as other models like linear regression, produced results with very little correction and low interpretability. It was determined that the dataset was too small for any model to effectively ”learn.” For this reason, it was decided to expand the dataset back to the original 4000 instances, including data from across the globe. Initially, there was a concern that this would overgeneralize the dataset and possibly create bias towards conditions in regions that had the most instances of wildfires present in the dataset. This concern was not forgotten, but previous results demonstrated that 435 rows was not enough data to create an effective predictive model.

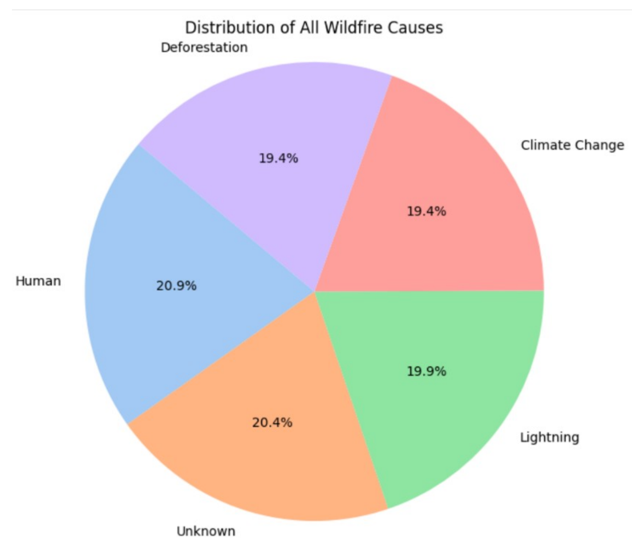

Fig. 4. Distribution of Causes of Wildfires Across all Wildfires

C. Logistic Regression

Logistic Regression was performed on the dataset with the globalized data to classify fires with a cause of ”Unknown” into the other four cause categories: Deforestation, Lightning, Climate Change, and Human. The data was biased by the classes in which the cause was known (4) vs the classes where the cause was unknown (just 1). As can be seen by the pie chart in Figure 4, that is roughly 80% vs 20%, leading to a natural bias towards known causes. In previous iterations of just predicting ”Unknown” causes the linear regression model was almost always able to predict when the cause was known, but rarely able to predict when the cause was unknown. To handle this class imbalance, scaling and SMOTE (Synthetic Minority Over-sampling Technique) were used. SMOTE uses machine learning to create synthetic instances of the minority class to more effectively train models. Additionally, making sure the data was trained on only known causes improved accuracy.

The results of the confusion matrix for logistic regression with a 70% tests size are shown in Figure 5. This yielded a 30% accuracy for the training model to predict a cause. The model predicted that 235 of the fires were due to climate change, 211 due to deforestation, 211 due to lightning, 171 due to humans.

Due to a relatively low accuracy from the initial logistic regression model, it was predicted that the synthetic data (i.e. fires before 1950) may be acting as noise in the dataset and making accurate predictions difficult for the model to accomplish. It was decided to reduce the dataset to just the non-synthetic data, so fires occurring after 1950 and after. The dataset was still 2000 rows and therefore, the size of the dataset being too small was not a concern, as it had been for analyzing just U.S. data.

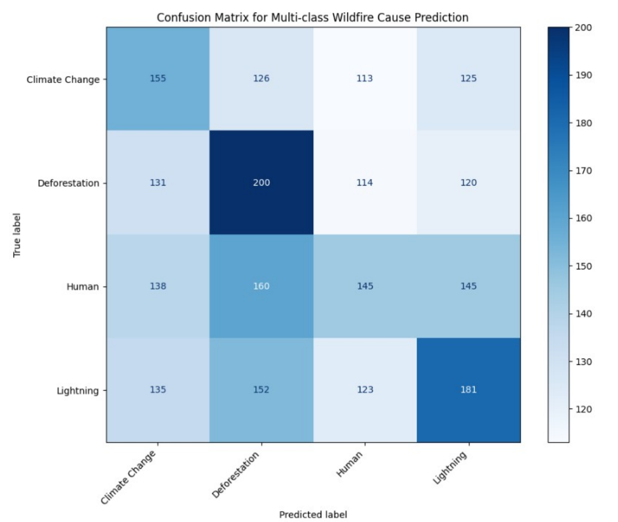

Fig. 5. Confusion Matrix for All Data

The results of the confusion matrix for linear regression with data before 1950 excluded, with a 70% tests size are shown in Figure 6. This yielded a 60% accuracy for the training model to predict a cause. The model predicted that 30 of the fires were caused by lightning, 28 by deforestation, 25 by climate change, and 24 by human causes.

D. Random Forest

The Random Forest method was performed on the dataset with the globalized data prior to the year 1950 to classify fires with their cause labeled as ”unknown.” Data accumulated from after 1950 was used because of the 1881-1950 synthetic data discovery. This model was first used on the full dataset, but results showed little to no correlation or relation within the dataset. When using data prior to 1950, the Random Forest model’s prediction showed more promising results. The model used a small testing group of 152 known causes, and 44 unknown causes. The overall accuracy of the model only reached 23% which showed some accuracy, but barely outperformed chance.

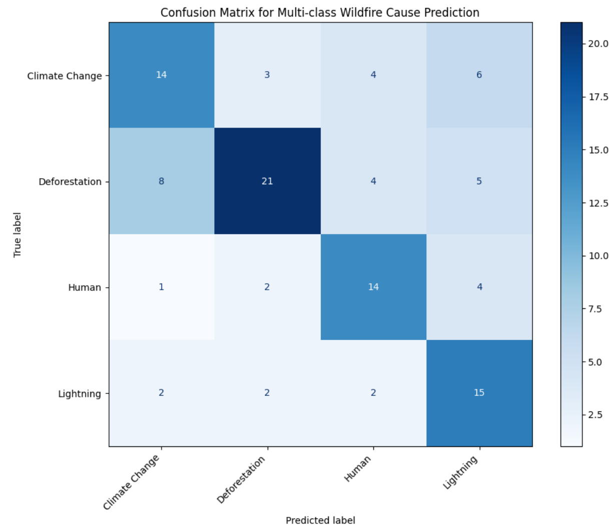

Fig. 6. Confusion Matrix for Wildfires in 1950 or After

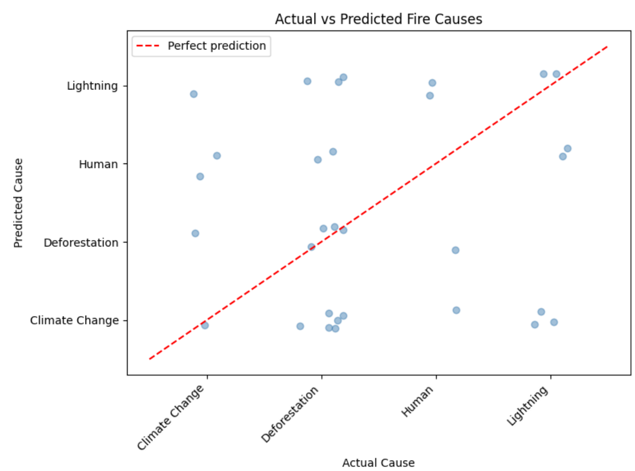

Fig. 7. Random Forest Scatterplot

Upon viewing Figure 7, only deforestation showed prediction reliability, while the other causes don’t show this similar trend. As shown in Figure 9, the prediction of deforestation was correct 2 out of 3 times and showed a 67% precision, compared to the other causes that showed only 0-25% precision. Using this model, fires caused by humans were never predicted successfully, resulting in a 0% precision. Every prediction has a confidence below 65% with most falling between 28-45%. The model is showing uncertainty on nearly every unknown wildfire. The Random Forest model did not produce a meaningful result and barely learned anything from our dataset values.

E. Support Vector Machine

Support Vector Machine (SVM) models were also used on the dataset as classification tools. For these models, the ”Country” attribute was completely removed, and ”Region” and ”Cause” were one-hot encoded. The dataset was also

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Known rows:   152
Unknown rows: 44

Cause classes: ['Climate Change' 'Deforestation' 'Human' 'Lightning']

Classification Report (on known causes):
                precision    recall  f1-score   support

Climate Change       0.09      0.20      0.12         5
 Deforestation       0.67      0.27      0.38        15
         Human       0.00      0.00      0.00         4
     Lightning       0.25      0.29      0.27         7

      accuracy                           0.23        31
     macro avg       0.25      0.19      0.19        31
  weighted avg       0.39      0.23      0.26        31
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Fig. 8. Random Forest Results

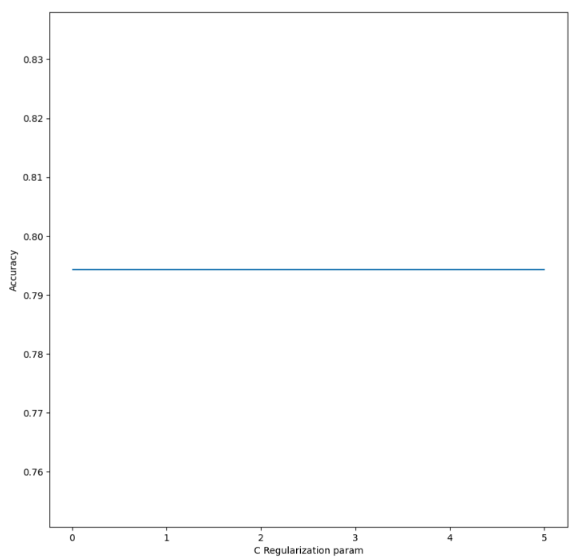

Fig. 9. C Regularization Parameter Optimization; accuracy stagnant at 0.7943

scaled using the Standard Scaler method in sklearn library. This analysis spanned data from the years containing non-synthetic data (1950-2025). Three types of kernels, linear, polynomial, and radial basis function, were evaluated for optimizing the C regularization parameter. This parameter is evaluated from 0.1 to 5 for each kernel. It is found that the accuracy is stagnant at approximately 0.7943 for each value of the C parameter, for each kernel. Figure **??** shows the C parameter vs. Accuracy plot for the linear kernel; both the polynomial and RBF kernels analyses produced the same plot.

F. Light GBM

The Light Gradient Boosting Machine (Light GBM) is a framework developed by Microsoft that builds complex tree-based predictive models. For this method, categorical data can remain in the dataset without any encoding methods. Numerical data can also be used without the need for scaling.

The implementation of a LightGBM model in Python is fairly

IV. Discussion

Logistic Regression yielded the highest accuracy of any model used at 60% when the synthetic data was removed. As can be seen in both Figures 5 and 5, the data appeared to be biased toward wildfires caused by deforestation. Both iterations demonstrated that the model struggled to predict wildfires caused by humans. This is likely because, as discussed in previous sections, there would be no discernible pattern in weather factors for fires caused by humans. Additionally, the attributes for the wildfires in this dataset do not consider any human factors, like distance from roads or population density in an area. These would be the most important factors used to predict human-caused fires.

Using the random forest model with this dataset produced only a 23% accuracy which proved this model was not the best option. However, when comparing these results to other studies, random forest models can be one of the best choices for wildfire datasets. To further test with this model, there may need to be more variables present. Studies that had the best performance of random forest models [11] [12] had other variables like vegetation, terrain, and socioeconomic data to get accurate results. Another factor that would have given more accurate predictions was the size of the dataset used. A study that has the random forest model as the second best performer emphasized the importance of "...using a large dataset which includes wildfire observations from around the globe over a full year" [13]. The dataset used for our project used locations across the world, but failed to provide data for wildfires throughout the entire year, ranging from only May to October. Shown in several studies, to get a reliable prediction there needs to be an increase in variables and size of the test group.

Some strategies are formulated for further tuning of SVM models on this dataset. Firstly, wider range of C parameter values could be investigated for higher accuracy. For the RBF kernel specifically, a technique like GridSearchCV could be used for tuning the gamma hyperparameter. Assuming the performed methodology of the SVM model is sound, this accuracy stagnation could be indicative of too much of an even split between each of the causes. It is also likely that the attributes of the dataset may be too unspecific for meaningful patterns to arise. Attributes like wind speed and humidity are averages per area per month, whereas the fire cause is the most frequent per month. A dataset containing individual instances of recorded fires, rather than averages and totals per month per region, could serve as better candidates for the classification models employed in this paper.

Logistic Regression has frequently been used for the prediction and explanation of human-caused fire occurrences [14] [15]. Logistic regression is likely well-suited to the analysis of non-linear factors, like those associated with humans, because it predicts the probability of a category. Logistic regression provides a medium alternative between more complex algorithms like machine learning and neural networks, and simpler models such as linear regression. Human causes associated with wildfires are the most difficult for models to predict, which is supported by the findings of this report and other research.

V. CONCLUSION

talk about combining models challenges of regions – why we had to make choice at beginning

The research surrounding Machine Learning algorithms to predict human-caused wildfires shows that Machine Learning algorithms improve the accuracy of logistic regression [17].

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